

RED SEAL EXAM PREP

308A Sheet Metal Worker

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
<https://redsealquiz.ca> · Edition 2026-07

About This Material

Independent, unofficial study resource. This publication is not affiliated with, endorsed by, or approved by the Canadian Council of Directors of Apprenticeship (CCDA), Employment and Social Development Canada, the Red Seal Program, Skilled Trades Ontario, or any provincial or territorial apprenticeship authority. "Red Seal" is used solely to describe the examination for which this material provides practice.

Originality. All questions and explanations in this publication are original works created by the author. This material references only publicly available occupational standards (Red Seal Occupational Standard) for topic alignment and reproduces no content from any actual examination.

No guarantee of results. This study material is provided for educational and informational purposes only. Success on any certification examination depends on many factors, and no representation is made that using this material will result in passing any exam. Exam content, structure, and weightings are set by certification authorities and may change without notice.

Technical accuracy. Code values and technical specifications cited in explanations reflect commonly referenced Canadian standards at the time of writing. Always defer to the current edition of the applicable code or the manufacturer's documentation in real work.

Refunds. Because this is an instantly delivered digital file, all sales are final once the download link has been delivered. If you hit a technical problem — a corrupted or inaccessible file, or a duplicate purchase — contact us within 14 days and we will repair the file or refund you in full.

Updates. Corrections and additions are published to the same file. Re-download from your Ko-fi library to get the current edition at no extra cost.

© 2026 the author. All rights reserved. For personal study use by the purchaser only — redistribution, resale, or file sharing is prohibited.

Mock Exam — answer all questions, then check the key

Q1. When developing a pattern for a cylindrical elbow using parallel line development, what determines the number of equal divisions the circumference is broken into?

- A) The number of gores or pieces making up the elbow
- B) The gauge of the sheet being formed and its bend allowance
- C) The centreline radius of the elbow divided by the duct diameter
- D) The accuracy required in the finished pattern curve

Q2. A sheet metal worker in British Columbia is shooting powder actuated fasteners into a concrete slab to hang duct. He has just loaded the tool when the foreman calls the crew off for a twenty-minute toolbox meeting. Under the OHS Regulation, what must he do with the tool?

- A) Unload it only if the interruption will run longer than one hour
- B) Unload it at once, because work after loading has been interrupted
- C) Leave it loaded, but lay it on the deck with the muzzle pointed down
- D) Leave it loaded and carry it along so no one else can pick it up

Q3. A length of round duct is cut off on a slant at one end. Seen square on, that cut opening is an ellipse. The piece is now developed by the parallel line method: the stretch-out is stepped off in equal divisions, and the height of the cut at each division is projected across from the elevation. What shape do the plotted heights trace along the cut edge of the pattern?

- A) A sine wave, one full cycle across the stretch-out
- B) A straight line, since the cut is made on a flat plane
- C) A circular arc struck from a single centre point
- D) An ellipse, matching the shape of the cut opening

Q4. Which welding process is most commonly used for joining stainless steel sheet metal in HVAC and food service applications?

- A) Flux-Core Arc Welding (FCAW) — self-shielding gives the cleanest stainless bead
- B) Oxy-acetylene welding — lowest heat input, so stainless never sensitizes
- C) Shielded Metal Arc Welding (SMAW/stick) — low spatter and best control on thin gauge
- D) Gas Tungsten Arc Welding (GTAW/TIG) — clean, precise welds on thin material

Q5. Before starting any work on an energised electrical panel in the mechanical room where ductwork is being installed, the sheet metal worker must:

- A) Apply LOTO procedures on all energy sources
- B) Wear rubber insulating gloves only
- C) Simply turn off the panel breaker and proceed
- D) Notify the foreman verbally and proceed

Q6. What is the primary purpose of a "break" or "hem" along the raw edge of a sheet metal panel?

- A) To stiffen the edge and remove the sharp hazard
- B) To provide a surface for sealant application
- C) To increase airflow velocity along the panel
- D) To allow for thermal expansion of the panel

Q7. Rectangular duct in 22 gauge is being joined with SMACNA transverse joint T-21, the welded flange, on a project whose specification adopts the SMACNA standards. Hangers are not being provided at every transverse joint. What do the notes for that joint then require at the duct corners?

- A) A 6.3 by 102 mm rod brazed or welded in the corners
- B) A 16 gauge corner piece closed with 9.5 mm bolts
- C) A continuous fillet weld run around every corner joint
- D) A gasket lapped at the corner and bolted through

Q8. Rectangular duct is being fabricated for the suction side of an air handling unit and is assigned the 3 in. w.g. (750 Pa) negative pressure class. Which statement about that duct agrees with the SMACNA duct construction standards?

- A) Its usual failure is buckling of the duct wall and reinforcements
- B) Its seams may be left unsealed, because the leakage is drawn inward
- C) It needs fewer wall fasteners, as suction holds the stiffeners on
- D) It calls for a heavier wall gauge than the same positive class does

Q9. A duct system shows higher than design static pressure drop across a filter bank. What is the most likely cause?

- A) Duct leakage downstream of the filter
- B) Supply fan speed set too low
- C) Excessive outdoor air intake volume
- D) Filters loaded with dust and dirt

Q10. What is the primary purpose of insulating HVAC supply ductwork in an unconditioned attic?

- A) To reduce duct velocity noise in the attic
- B) To prevent condensation and reduce thermal loss
- C) To meet SMACNA pressure class requirements
- D) To allow higher static pressure ratings

Q11. A sheet metal worker is applying flashing near an unguarded roof edge. How does a travel restraint system differ from a fall arrest system?

- A) Restraint prevents reaching the edge; arrest stops a fall
- B) Restraint suits sloped roofs; arrest suits only flat ones
- C) Restraint needs no anchor; arrest needs a rated anchor
- D) Restraint requires a full harness; arrest allows a belt

Q12. In British Columbia, a sheet metal worker is cutting on a layout table when a sheet slips and lacerates his forearm. He is treated at a walk-in clinic, where the wound is closed with sutures, and he returns to finish his shift. Under the Workers Compensation Act, the employer must:

- A) Report the injury; no investigation is required
- B) Report the injury and investigate the incident
- C) Investigate it, but the clinic reports the injury
- D) Record it as first aid, since no time was lost

Q13. A commercial building has a large mechanical exhaust system, and a makeup air unit is installed to replace the air it removes. Under Article 6.3.2.8. of Division B of the National Building Code of Canada 2020, how must the makeup air facilities be arranged?

- A) Switched on by hand at the panel whenever exhaust is running
- B) Left as fixed louvres, with no damper, fan or control on them
- C) Sized to deliver ten percent more air than the exhaust removes
- D) Interlocked with the exhaust devices so both operate together

Q14. A 600 mm diameter round main is to be tapped for a branch using a saddle fitting. Under the SMACNA round duct construction standards, what is the largest branch diameter that connection may serve?

- A) 500 mm
- B) 400 mm
- C) 450 mm
- D) 300 mm

Q15. A rooftop exhaust fan is shut down overnight and outdoor air blows back down the duct into the building. Which unpowered damper is fitted in the duct to stop this?

- A) A manual volume control damper
- B) A motorised isolation damper
- C) A fire damper with a fusible link
- D) A backdraft (gravity) damper

Q16. Portable welding screens are being set up around an arc welding station in an occupied fabrication shop so that other workers in the bay are not exposed to the arc. Under the BC Occupational Health and Safety Regulation, what must those screens themselves be?

- A) Opaque, and bonded into the shop grounding system
- B) Flame resistant, with a nonreflective surface finish
- C) Tinted to a welding shade so the arc stays visible
- D) Rigid steel panels, at least two metres in height

Q17. ASHRAE 62.1 governs ventilation for acceptable indoor air quality. Which parameter does it specify for commercial occupancies?

- A) Minimum refrigerant charge weight per equipment room
- B) Maximum permitted duct velocity in metres per second
- C) Minimum outdoor air rate per person and floor area
- D) Maximum allowable CO₂ concentration in the flue gas

Q18. A panel is being run through a Pittsburgh lock former to form the pocket on one edge. How is the panel presented to the machine?

- A) Lying flat, the edge fed squarely into the rolls against the guide
- B) Standing on edge in the rolls; panel orientation makes no difference
- C) Entered from the outfeed side so the pocket runs toward the operator
- D) Lying flat, but entered at about 45 degrees to the line of the rolls

Q19. When developing the pattern for a right cone using radial line development, what measurement represents the radius of the arc drawn in the pattern?

- A) The diameter of the cone base
- B) The slant height of the cone
- C) The radius of the cone base
- D) The vertical height of the cone

Q20. Brazing differs from welding in that:

- A) Brazing requires flux; welding never does
- B) Brazing is used only for aluminium work
- C) Brazing melts the base metal; welding does not
- D) The filler melts but the base metal does not

Q21. A rooftop outdoor air intake is being located for a commercial air handling unit. The nearest source of contaminants is the kitchen cooking exhaust outlet of the restaurant below. Under Table 6.3.2.9. of Division B of the National Building Code of Canada 2020, what is the minimum distance required between them?

- A) 3.0 m
- B) 1.5 m
- C) 3.5 m
- D) 7.6 m

Q22. An outdoor air intake hood and an exhaust outlet are being fabricated for a rooftop unit. Under Article 6.3.2.9. of Division B of the National Building Code of Canada 2020, the corrosion-resistant screen fitted at each of those exterior openings must have openings no larger than:

- A) 25 mm
- B) 50 mm
- C) 15 mm
- D) 6 mm

Q23. A branch is to be cut into an existing 450 mm diameter round main and connected with a saddle fitting. Under the SMACNA round duct standards, what governs that connection?

- A) Branch up to two thirds of the main, with nothing projecting in
- B) Any branch size, once the saddle is screwed and then sealed
- C) Branch up to half the main, the collar finishing flush inside
- D) Branch up to three quarters of the main, a short stub inside

Q24. In SMACNA's notes for the transverse joints of Figure 1-4, one of these joints is held to a 2 in. w.g. maximum static pressure. Which one?

- A) Flat S slip
- B) Standing S slip
- C) Welded flange
- D) Companion angle flange

Q25. A long seam welded as one continuous pass in 20 ga sheet has left the panel buckled between the tacks. Which change best controls the distortion?

- A) Weld short stitches along the seam, backed by chill bars
- B) Open the root gap so the joint has more room to move as it heats
- C) Run one continuous pass at a lower travel speed to even the heat
- D) Preheat the whole panel before welding so it expands evenly

Q26. What is a Pittsburgh lock seam and for what application is it primarily used?

- A) A spiral seam used on round ductwork formed by a spiral pipe machine
- B) A cross-seam used to connect two sections of rectangular duct end-to-end
- C) A double-folded flat seam used on architectural metal roofing
- D) A longitudinal seam used to close the sides of rectangular duct sections

Q27. A large rectangular duct is being built with internal ties, and the fabricator wants to use 25 mm by 3.2 mm galvanized steel tie straps rather than rod. Which statement about that choice agrees with the SMACNA duct construction standards?

- A) They suit either pressure mode once sized from the load table
- B) They are for positive pressure only, set edge-on to the flow
- C) They may be plain uncoated steel inside a galvanized duct
- D) They remove the need to attach reinforcement to the wall

Q28. A sheet metal contractor is running the exhaust duct from laundry-drying equipment in a multi-storey building. Under Article 6.3.2.10. of Division B of the National Building Code of Canada 2020, what does the Code require of that duct?

- A) Independent of other exhaust ducts, cleanable, smooth-walled
- B) Tied into the washroom exhaust at the inlet of a common fan
- C) Run in flexible corrugated duct, discharging to a service shaft
- D) Insulated and discharged into a ventilated attic or roof space

Q29. What is the "seam line" on a developed pattern?

- A) The reference line from which all element lengths are measured
- B) The outer boundary of the developed flat blank, followed when cutting
- C) The line the blank is folded on to form a corner of the fitting
- D) The line where the pattern is joined to form the fitting

Q30. When working in a ceiling plenum space above a suspended ceiling, what is a key safety concern specific to sheet metal workers?

- A) Energised conduit contact and falls through the grid
- B) UV exposure from the fluorescent and LED fixtures overhead
- C) Sustained noise above 85 dBA from the operating HVAC equipment
- D) Chemical burns from duct sealant curing in the enclosed space

Q31. What is the purpose of a "notcher" in sheet metal fabrication?

- A) Punching rivet holes at set spacing intervals
- B) Forming Pittsburgh lock seams on duct sections
- C) Cutting square notches at blank corners for bending
- D) Cutting circular holes for duct collars

Q32. When using a slip roll to form a cone (tapered cylinder), how is the taper achieved?

- A) One end of the rear roller is raised
- B) One end of the front pinch rolls is set tighter than the other
- C) The sheet is fed at an angle using a guide fence
- D) A tapered forming die is inserted between the rolls

Q33. When a fitting has both curved and flat surfaces (e.g., a round-end rectangular plenum takeoff), what development approach is used?

- A) Radial line development for every surface
- B) The curved section is approximated as flat — no special method needed
- C) Triangulation for all surfaces regardless of shape
- D) Parallel line for flat sides, triangulation for the transition

Q34. A rectangular duct 600 mm x 400 mm is fabricated from unlined 20 ga (1.0 mm) galvanised sheet for a 2 in. w.g. (500 Pa) system, with transverse joints at 2.4 m (8 ft) spacing. Applying the SMACNA duct construction standards, what is required of the flat duct sides?

- A) The run must be broken into sections shorter than 1.5 m
- B) Crossbreaking may replace the required reinforcement
- C) The 600 mm sides must be crossbroken or beaded
- D) The 400 mm sides must be crossbroken or beaded

Q35. A round-to-round eccentric reducer is to be developed. Its apex is not centred over the two openings. Which statement is correct?

- A) Radial line development applies directly, as it is still a right cone
- B) Each element line must be found in true length, as they differ
- C) Parallel line development applies, since both openings are round
- D) A concentric reducer pattern will do if the small end is cut on a slant

Q36. A 205 L steel drum that previously held a solvent-based duct sealant is to have a lifting bracket welded to its side. Under the BC Occupational Health and Safety Regulation, what has to happen before the arc is struck?

- A) Airing the drum open for a full day before the work
- B) Opening the bung and keeping an extinguisher at hand
- C) Thorough cleaning, then testing by a qualified person
- D) Purging the drum with oxygen and leaving it vented

Q37. SMACNA's notes for longitudinal seams state a maximum static pressure for the button punch snap lock seam. What is that limit?

- A) 10 in. w.g.
- B) 1 in. w.g.
- C) 2 in. w.g.
- D) 4 in. w.g.

Q38. A shop is making slip-and-drive transverse joints on rectangular duct that has to be sealed to a specified sealing class. SMACNA warns that two things about the connection must be controlled or the sealing will be compromised. What are they?

- A) Rivet spacing and the size of the rivet head used
- B) Sheet gauge and the pressure class of the section
- C) Sealant brand and the temperature at application
- D) Connector length and notch depth at the duct ends

Q39. A sheet metal worker is spot welding two 20ga mild steel panels. What parameter most directly controls nugget size?

- A) Electrode material (copper alloy vs carbon)
- B) Number of spots per metre
- C) Weld current (amperage)
- D) Panel orientation (horizontal vs vertical)

Q40. A sheet metal worker pours solvent-based duct sealant from the supplier's labelled pail into a smaller shop can. The can will sit on the bench all week for anyone on the crew to use. Under WHMIS, what must appear on that can?

- A) Product name only, since the can stays in the shop for the crew to use
- B) Product name, safe handling precautions and a safety data sheet reference
- C) The supplier hazard pictogram and signal word copied from the pail label
- D) Product name, the supplier address and phone number, and a signal word

Q41. A rectangular duct section 400 mm by 250 mm is to be made from one sheet, closed with a Pittsburgh lock at one corner. What width of flat blank does the layout call for?

- A) 1300 mm, the girth already covering the lock
- B) 1300 mm, with the lock allowance taken out of it
- C) 1300 mm, plus the pocket and edge allowances
- D) 650 mm, plus the pocket and edge allowances

Q42. A sheet metal worker is going to dress the nicked edge of a cold chisel on the shop's pedestal grinder. Before the wheel is started, how must the work rest be set?

- A) About 15 mm off the wheel, its edge at or above wheel centre
- B) Within 3 mm of the wheel, its edge at or above wheel centre
- C) Swung clear of the wheel, so long work can be laid on flat
- D) Within 3 mm of the wheel, its edge set below wheel centre

Q43. A duct fitter is closing a fold by hand on 24 gauge galvanized sheet. Why is a rawhide, hide or plastic mallet used on the sheet rather than a steel hammer?

- A) A steel face stretches metal and cuts the coating
- B) A steel face would spark near the duct sealant
- C) Steel hammers are barred from coated sheet work
- D) A mallet is heavier, so fewer blows are needed

Q44. On a true-length diagram for a triangulated transition, an element line has a plan (horizontal) length of 400 mm and a vertical height (rise) of 300 mm. What is its true length?

- A) 250 mm
- B) 350 mm
- C) 700 mm
- D) 500 mm

Q45. A VAV (Variable Air Volume) system modulates airflow to individual zones. What controls the airflow at each zone terminal?

- A) Manual balancing dampers set at commissioning
- B) A VAV box with a thermostat-controlled damper
- C) A fixed orifice plate in the branch duct
- D) The AHU supply fan speed alone

Q46. What is the purpose of a turning vane installed inside a 90° rectangular duct elbow?

- A) To split the airflow evenly between two branches taking off at the elbow, balancing the system
- B) To stiffen the elbow corners so a lighter gauge of sheet can be used at the bend
- C) To guide airflow smoothly around the bend, reducing turbulence and noise
- D) To break the airstream into layers so condensation cannot form on the elbow's inner wall

Q47. A grooved seam is commonly used in which application?

- A) As a standing seam on architectural metal roofing
- B) As a transverse (cross) connection between rectangular duct sections
- C) As the spiral seam on round duct and on joined curved edges
- D) As the longitudinal seam on rectangular duct sections

Q48. A box and pan brake is used for what sheet metal forming operation?

- A) Forming the Pittsburgh lock seam on duct sections and corners
- B) Cutting sheet metal blanks to exact length before any forming
- C) Rolling flat sheet into cylinders and round duct sections
- D) Bending sheet metal into boxes and pans at precise angles

Q49. When installing solid rivets to join two sheets of galvanized steel, what preparation is required before driving the rivet?

- A) No hole is needed — solid rivets self-pierce the sheet metal as they are driven
- B) Heat the rivet to red heat before inserting so the shank shrinks tight as it cools
- C) Drill a hole to size, insert the rivet, and buck the head while driving
- D) Apply acid flux around the hole first so the rivet seals itself against galvanic corrosion

Q50. What is "springback" in sheet metal bending and how is it compensated?

- A) Thermal expansion during welding — corrected by pre-heating
- B) Stretching at the bend — corrected by adding material
- C) Sheet vibration during cutting — corrected by clamping
- D) Elastic return after bending — corrected by overbending

Answer Key & Explanations

Q1. Answer: D

In parallel line development the plan view of the duct is divided into equal parts and those same divisions are then stepped off along the stretch-out line, so the circumference divisions and the stretch-out divisions are one and the same set and neither can determine the other. What sets the number is how closely the pattern has to follow the true curve. Each division produces one point where the mitre line is projected across onto the stretch-out, and the pattern outline is drawn by joining those points, so a coarse division cuts corners off the curve while a fine one traces it closely at the cost of layout time. Twelve or sixteen equal parts is the usual shop compromise. The Red Seal Occupational Standard states the criterion outright in sub-task B-6.02: the plan and elevation are divided into equal parts to achieve required accuracy. The number of gores decides how many separate patterns an elbow needs, not how finely each one is divided. Sheet gauge and bend allowance change the flat blank size and the seam and edge allowances, not the spacing of the division lines. Centreline radius divided by diameter is the elbow throat ratio, a fitting-selection figure with nothing to do with layout spacing.

Q2. Answer: B

Section 12.56(5) of the BC OHS Regulation is unconditional: a powder actuated tool may only be loaded when it is being prepared for immediate use, and it must be unloaded at once if work is interrupted after loading. A toolbox meeting is an interruption, so the power load comes out before he walks away. Section 12.55(1) then takes over for the tool that is not in use, requiring it to be unloaded and the tool and the power loads to be securely stored and accessible only to qualified and authorized persons, and section 12.56(1) limits handling and use of the tool and the loads to a qualified person. The tempting answer is to leave it loaded with the muzzle pointed at the deck. Muzzle discipline is sound practice and it is what a worker does while the tool is in use, but the Regulation does not offer it as a substitute for unloading, and a loaded tool set down on a deck is exactly what the storage rule exists to prevent, because the next person to pick it up may not be a qualified person. There is no time threshold in the Regulation that would let a short interruption pass, and carrying the loaded tool to the meeting keeps a charged tool live among a crowd of workers, which is the same violation with the tool in his hands.

Q3. Answer: A

Two different curves are in play and the item turns on keeping them apart. The cut face, the opening seen square on, is a plane section of a cylinder and is an ellipse. The cut edge on the flat pattern is a different curve, because developing the piece lays the circumference out straight while leaving every height as it was. Why it comes out as a wave is visible in the elevation. The mitre line there is straight, and each height is read where that straight line crosses an element. But the elements do not project evenly onto the elevation: they crowd together toward the two outside edges of the pipe and open out through the middle, so a point standing at an angle θ around the pipe projects to a horizontal position proportional to the cosine of θ , and the straight mitre line therefore hands it a height proportional to the cosine of θ as well. On the stretch-out those same points are spaced evenly, one equal division apart. Replotting cosine heights at even spacings is exactly what turns the straight line into a wave: measured from the mid-height of the cut, the height is the radius times the tangent of the cut angle times the cosine of the angle travelled around the pipe, plotted against arc length. That is a sinusoid, and it runs one full cycle over the stretch-out, with a crest at the longest element, the mid-height crossed at the two side elements, and a trough at the shortest element. Whether it reads as a sine or a cosine depends only on which element the stretch-out is started from; the curve is the same. An ellipse answers a different question. It is the true shape of the hole, and it is what the cut looks like while the metal is still round. Setting an ellipse out on the blank instead gives a piece whose end will not lie in a plane once it is rolled up. No circular arc will fit the curve, whatever radius is chosen, because an arc bends the same way from end to end. This curve does not: it arches over the longest element and hollows under the shortest, reversing at the two side elements. That is why the layout is finished by joining the plotted points with a flexible curve rather than by swinging an arc from a centre. A straight line would be right for a prism, not a cylinder. Rectangular duct cut on a slant does develop with straight pattern edges, one straight run across each face, and that is where the expectation comes from. On round duct the mitre line is straight only in the elevation, where the elements are foreshortened; once the even stretch-out spacings replace those foreshortened ones, the line becomes a wave.

Q4. Answer: D

GTAW (TIG welding) is the preferred process for stainless steel sheet metal because it provides precise heat control, produces clean welds without spatter, and maintains the corrosion-resistant properties of stainless steel. SMAW and FCAW create too much spatter and heat distortion on thin gauge material. Oxy-acetylene may cause carbide precipitation in stainless steel.

Q5. Answer: A

LOTO (Lockout/Tagout) is mandatory before working near or on energised equipment. The sheet metal worker must de-energise, lock out, and verify zero energy state.

Q6. Answer: A

A hem folds the raw cut edge back on itself, removing the sharp edge hazard and adding rigidity to the panel edge.

Q7. Answer: A

The notes for Figure 1-4, joint T-21, welded flange, call for a 5/8 inch, that is 16 mm, minimum flange with an edge weld, and add that on 24, 22 and 20 gauge the fabricator is to braze or weld a 1/4 inch by 4 inch rod, 6.3 by 102 mm, in the corners or else provide hangers at every joint. The reason sits in how this joint carries load. On light gauge there is no angle and no bolted frame; the turned-up flange and its edge weld are the whole joint, and the corner is where two flanges meet at right angles and where the weld tears first when the duct is hung between joints and the section works. A short rod brazed or welded across each corner ties the two flange legs together and puts metal where the tearing starts, and the alternative the note allows is to take the load off the joint entirely by hanging at every one of them. The 16 gauge corner piece with 9.5 mm bolts is the near miss and a genuine SMACNA requirement, but it belongs to a different joint, T-24, the formed flange, where mating flanges are rolled on the duct ends to form a tee and steel corner pieces with 3/8 inch minimum bolts close the corners. Reading a T-24 corner rule onto a T-21 welded flange is the mistake this question is built on. Running a continuous fillet weld around every corner puts a great deal of heat into thin sheet, distorts the flange and is not what the note asks for. A gasket seals a joint and adds no strength at all, and the corner problem here is structural. Note also the related T-21 rule that flanges larger than 16 mm must be spot welded, bolted, riveted or screwed as required to prevent separation and weld breakage.

Q8. Answer: A

The pressure mode changes how a duct behaves, not what it is built from. SMACNA's notes on specimen testing set the two modes side by side: typical failures under positive pressure are joint separation from the duct, opening of joints, tearing of joint corners, screw or rivet pullout from the duct near corners, and longitudinal seam separation, while typical failures under negative pressure are buckling of the duct wall at the corners, buckling of joints and reinforcements near the centre, openings that change in size or orientation enough to spoil the seal, fastener release, and excessive deflection. The same notes add that a negative-mode test is more critical for sheet deflection and for joint and stiffener buckling, that a test in one mode will not substitute for one in the other, and that the pressure capacity of a duct is usually higher in the positive mode than in the negative. Suction is the mode that collapses duct inward, so buckling of the wall and of the members stiffening it is what the fabricator is guarding against. The wall gauge is not where the difference shows up on rectangular duct: every rectangular duct reinforcement schedule, Tables 1-3 through 1-9, is headed for its class as w.g. static, pos. or neg., so one schedule serves both modes and this duct takes the same gauge and reinforcement grade as a 3 in. w.g. positive duct of the same size and reinforcement spacing. Round duct is the case that does differ, having a separate negative pressure gauge table of its own, but this duct is rectangular. Sealing is not relaxed on the suction side either. Section S1.9, which defines what a requirement to seal means, states that the requirements to seal apply to both positive and negative pressure modes of operation, so this duct still takes the sealing class that Table 1-2 assigns to its pressure class, and at 3 in. w.g. that is all transverse joints and all longitudinal seams. Fasteners go the opposite way to the guess: suction pulls the duct wall away from an external stiffener rather than pressing it on, and the introduction to the rectangular schedules states that negative pressure reinforcement attachments to the duct wall are generally at closer intervals than on positive pressure service, with the commentary adding that a reinforcing member's ability to do its job is critically affected by the location and adequacy of its attachment and that those variables change between the two modes. SMACNA is a construction standard rather than Canadian law; it applies on Canadian work where the project specification or an adopting code calls it up.

Q9. Answer: D

A clogged/loaded filter has high resistance, causing elevated pressure drop across the filter section. This also reduces airflow. Filters must be replaced or cleaned.

Q10. Answer: B

Insulation on supply ducts in unconditioned spaces prevents condensation (cold ducts in humid attics), reduces heat gain from the attic, and improves system efficiency.

Q11. Answer: A

The two systems answer different questions. A travel restraint system is rigged so the worker physically cannot get to the edge: the lanyard is short enough that the fall never happens. A fall arrest system assumes the worker will go over and is built to stop the fall before they strike anything below. That is why the regulations rank restraint ahead of arrest. The Ontario construction regulation requires a guardrail system first, and where that is not practicable, the highest ranked practicable method from travel restraint, then a fall restricting system, then fall arrest, then a safety net. Roof slope does not decide which system is used. Both must be attached to a fixed support, so a restraint lanyard is anchored just as an arrest lanyard is. The hardware difference runs opposite to the common belief: travel restraint may use a full body harness or a safety belt, while fall arrest requires a full body harness with a shock-absorbing lanyard, because arresting a fall in a belt injures the worker.

Q12. Answer: B

What decides this is medical treatment, not lost time. A nick dressed at the shop and entered in the first aid record goes no further, but a wound closed with sutures at a clinic is health care, and it engages two separate employer duties that run alongside each other. On reporting: section 150(1) of the Workers Compensation Act requires an employer to report to the Board, WorkSafeBC, within three days after its occurrence, every injury to a worker that is or is claimed to be an injury arising out of and in the course of the worker's employment. That duty does not wait to see whether a shift is lost, and it is the employer's own - whatever report a clinic or an attending physician makes on its own account, section 150(6) makes an employer who fails to report guilty of an offence. Section 150(7)(a) allows the Board to except a category of minor injuries from reporting by regulation, which is the space a first aid entry occupies, but the Act sets the investigation threshold at injury requiring medical treatment, and closing a wound with sutures is medical treatment. On investigating: section 69(1)(b) puts an incident that resulted in injury to a worker requiring medical treatment among those the employer must investigate. Section 71 requires the preliminary investigation immediately, with its report completed within 48 hours, and section 72 the full investigation, whose report goes to the Board within 30 days. Nothing in that turns on the injured worker asking for it. Other provinces impose their own versions of both duties, on their own bodies and their own time limits, which is why the scenario names a province.

Q13. Answer: D

Article 6.3.2.8. has three sentences and they build on one another. Sentence (1) requires that in ventilating systems that exhaust air to the outdoors, provision be made for the admission of a supply of makeup air in sufficient quantity so that the operation of the exhaust system and other exhaust equipment or combustion equipment is not adversely affected. Sentence (2) then settles how: makeup air facilities required by Sentence (1) shall be interlocked with the exhaust devices they serve so that both operate together. Sentence (3) adds that where makeup air is introduced directly from outdoors to occupied parts of the building in winter, it shall incorporate means of tempering that air to maintain the indoor design temperature. The interlock is the point of the Article. What the Code is guarding against is depressurisation: run a big exhaust fan with nothing feeding the building and the room goes negative, the exhaust falls short of its rated flow, and a naturally drafted fuel-fired appliance in the same space can spill its combustion products back into the room. Hand switching fails the requirement because it depends on somebody remembering, and it is exactly on the day nobody remembers that the building is depressurised. A deliberate ten percent over-supply is a pressurisation choice a designer may make for a clean room or a vestibule, not a Code rule, and it does nothing about the exhaust running alone. Fixed louvres with no control cannot be interlocked with anything, and in a Canadian winter an untempered opening of that size delivers outdoor air straight into the occupied space, which Sentence (3) will not accept.

Q14. Answer: B

The round duct construction standards accept either a saddle or a direct connection of a branch into a larger duct, but they cap the branch: its diameter shall not exceed two-thirds of the diameter of the main. Two-thirds of 600 mm is 400 mm, so a 400 mm branch is the largest the connection may serve, and anything above it calls for a proper tee or a reducing fitting rather than a tap. Half the main diameter is the near miss because it is a widely repeated shop rule of thumb, and it is a conservative one, but it is not what the standard says and it will have the shop refusing a tap the standard allows. The same clause carries two conditions that matter as much as the ratio. Protrusions into the interior of the main are not allowed, so the branch is cut and fitted flush with the inside surface of the main rather than pushed through it. And all saddle fittings shall be sealed at all pressures, which is stricter than the sealing schedule the pressure class would otherwise call for, because a saddle sits on a curved surface and never closes tightly on its own. A direct connection must also include mechanical attachment sufficient to maintain the integrity of the assembly. The branch is not required to be heavier metal than the main: fittings take a wall thickness not less than that specified for longitudinal-seam straight duct. SMACNA is a construction standard rather than Canadian law and governs where the project specification or the authority having jurisdiction adopts it.

Q15. Answer: D

A backdraft damper, also called a gravity damper, has light blades hung on a horizontal axis and held at or near the closed position by their own weight, sometimes with a small counterweight to set how easily they lift. The fan's airflow pushes them open in the direction of flow, and when the fan stops they fall shut, so air cannot travel back down the duct. Nothing drives them - no actuator, no wiring, no interlock with the fan - which is the condition the stem sets. A motorised isolation damper will also shut an idle duct, and on a large system it often does, but it needs power and a control signal, so it does not answer a question about an unpowered device. A fire damper sits open in normal service and closes only when its fusible link releases at its rated temperature; it protects a fire separation and takes no notice of which way the air is going. A manual volume control damper is set by hand during balancing and locked in that position, so it neither opens on forward flow nor closes against reverse flow. All four are damper types a sheet metal worker fabricates; what separates them here is how each one is operated.

Q16. Answer: B

BC OHS Regulation 12.122(2) reads: "A screen, curtain or partition near an arc welding operation must be made of or be treated with a flame resistant material or coating, and must have a nonreflective surface finish." Both properties are there for a reason. Flame resistance is needed because spatter and hot slag land on the screen all day, and a curtain that will carry a flame turns a routine weld into a shop fire. The nonreflective finish is the requirement people forget, and it is the one that decides whether the screen does its job: arc ultraviolet bounces off a glossy or light, shiny surface, so a reflective screen can throw radiation onto the very workers standing behind it and give them the arc eye the screen was hung to prevent. Tinting the screen to a welding shade is the near miss, because that is exactly what the operator's helmet lens does and it sounds like the same problem. It is not. The operator has to see the puddle through the filter, so the lens must pass a controlled amount of light; a screen exists to keep radiation off people who are not looking at the weld at all, and it does not have to be seen through. A tinted panel with a shiny face would still reflect ultraviolet around the bay. Bonding the screen into the grounding system confuses the radiation control with the welding circuit, whose return path is the work lead clamped to the workpiece. Nothing in the section prescribes steel or a height; screens, curtains and partitions are all accepted, so long as they are flame resistant and nonreflective. Note also that 12.122(1) lets suitable eye protection for the exposed workers stand as the alternative to screening them.

Q17. Answer: C

ASHRAE 62.1 sets minimum outdoor air ventilation rates by occupancy type, expressed as a rate per person plus a rate per unit of floor area. That is the Ventilation Rate Procedure in Section 6.2: $Vbz = R_p \times P_z + R_a \times A_z$, where R_p (per person) and R_a (per unit floor area) are read from Table 6.2.2.1 in L/s or CFM. **It carries weight in Canada only because the Code adopts it: Sentence 6.3.1.1.(2) of Division B of the 2024 Ontario Building Code requires outdoor air to be supplied in accordance with Section 6.2 (Ventilation Rate Procedure), Section 6.3 (Indoor Air Quality Procedure) or Section 6.4 (Natural Ventilation Procedure) of ANSI/ASHRAE 62.1, and Sentence 6.3.1.1.(3) requires exhaust ventilation in accordance with Section 6.5.** In the 2012 OBC the same requirement sat at Sentence 6.2.2.1.(2), so older references and study notes will quote that number. Always use the edition your provincial code references, not the newest ASHRAE printing: the 2024 OBC and the National Building Code 2020 reference ASHRAE 62.1-2016, while the 2012 OBC referenced 62.1-2010. Because the 2016 edition dropped dwelling units from its scope, residential ventilation is covered by ASHRAE 62.2 or by Subsection 9.32.3 and CAN/CSA-F326, not by 62.1. The distractors belong to other documents entirely: duct velocity is a design and noise matter, refrigerant charge limits come from ASHRAE 15 and CSA B52, and flue gas CO₂ is a combustion analysis reading. None of the three appears in 62.1.

Q18. Answer: A

A Pittsburgh lock former does not fold the pocket in one action. The edge passes through a train of roll sets, each bending it a little further than the last, so the pocket is formed progressively along the whole length as the panel travels through. That is why presentation matters. The panel lies flat on the machine table and the edge to be formed is fed straight into the first roll set and kept tight against the guide, so the same amount of metal enters every pass and the finished pocket is the same depth from one end of the panel to the other. Start the edge in at an angle and the amount of metal in the rolls changes as the panel goes through, which tapers the pocket and can jam or spring the roll set. Standing the panel on edge takes the flat off the table and leaves nothing keeping the panel travelling straight, so orientation is very much not arbitrary. And the rolls are driven one way only, from infeed to outfeed, so a panel cannot be entered from the outfeed side at all.

Q19. Answer: B

In radial line development of a cone, all surface lines radiate from the apex. The radius of the arc in the flat pattern equals the slant height — the true length from the apex to the edge of the base circle. The arc length then equals the circumference of the base ($2\pi r$). The base radius fixes the included angle of that arc, not its radius, and the vertical height alone is always shorter than the slant height.

Q20. Answer: D

Brazing: filler melts (liquidus $>450^{\circ}\text{C}/840^{\circ}\text{F}$) but base metal remains solid. Welding: base metal melts. The key distinction is whether the base metal is melted.

Q21. Answer: A

Table 6.3.2.9., Minimum Distances of Air Intakes from Sources of Contaminants, gives a separate figure for each named source, and kitchen cooking exhaust is listed at 3.0 m. A vent for combustion products carries the same 3.0 m. The near miss is 3.5 m, which belongs to a sanitary vent, not to cooking exhaust; the two sit next to each other in the table and are easy to swap. The other figures in the same table are 1.5 m for a driveway, street or parking space, 4.5 m for the entry of a garage for five or more motor vehicles and for a garbage storage or pick-up area, and 7.6 m for a truck loading area or dock and bus parking, for a thoroughfare, arterial road, freeway or highway, and for the discharge of an evaporative heat rejection system. Two further sentences of the same Article govern the same intake. Sentence (3) requires outdoor air intakes to be installed not less than 0.3 m above roofs, landscape grades or other surfaces, taking anticipated snow accumulation into account, so on a Canadian roof the intake often has to stand well above that minimum. Sentence (2) also ties the location back to air quality, requiring the intake to be placed so the air entering the building meets the Code's quality provisions, which is the catch-all when the actual contaminant source is not one of the nine listed.

Q22. Answer: C

Sentence 6.3.2.9.(4) requires exterior openings for outdoor air intakes and exhaust outlets to be shielded from the entry of snow and rain and to be fitted with corrosion-resistant screens of mesh having openings not larger than 15 mm, except where experience has shown that climatic conditions require larger openings to prevent the screen openings from icing over. That exception is written for Canadian winters and is the reason a finer mesh is the wrong answer rather than a safer one: a 6 mm mesh has less free area, blinds off faster with snow and frost, and once it ices the intake is closed. Sentence (5) then requires the screens to be accessible for maintenance, which is what makes the hood's screen removable rather than riveted in place. Do not confuse this with Sentence (1) of the same Article, which uses the same number for a different purpose: supply, return and exhaust air openings located less than 2 m above the floor inside a room are protected by grilles whose openings will not pass a 15 mm diameter sphere. One rule is about weather, birds and debris at the building envelope; the other is about fingers and objects inside occupied rooms.

Q23. Answer: A

SMACNA S3.4 accepts both the saddle and the direct connection of a branch into a larger duct, then puts three conditions on them. The diameter of the branch shall not exceed two thirds of the diameter of the main, so on this 450 mm main the branch may go to 300 mm and no further. Protrusions into the interior of the main are not allowed, which rules out the stub of collar left standing inside that a hurried tap-in produces. And a direct connection into a main shall include mechanical attachment sufficient to maintain the integrity of the assembly. Sealing is the fourth line of the same clause and it is the one that catches people: all saddle fittings shall be sealed at all pressures. That is unconditional, so a saddle in a low pressure system where the designer has relaxed sealing elsewhere still gets sealed. Half the main diameter is stricter than the standard asks and would send perfectly good work back for no reason. A short stub inside is the misconception that a projecting collar makes a stronger tap - it does the opposite, adding a loss coefficient the designer never allowed for and giving dirt and lint somewhere to hang up. Screwing and sealing an oversized saddle does not rescue it: the two thirds limit is a limit on the geometry, and a branch approaching the diameter of the main removes so much metal from the main wall that no amount of fastening restores it. SMACNA is a construction standard rather than Canadian law and applies where a specification or an adopting code calls it up.

Q24. Answer: A

SMACNA's notes for Figure 1-4 give joint T-5, T-6, flat S slips, a line of their own: the slip is 24 gauge minimum and no lighter than two gauges below the duct, and when it is used on all four sides it is fastened within 2 in. of the corners and at 12 in. maximum intervals, at 2 in. w.g. (500 Pa) maximum static pressure. No other joint offered here is held that low. The standing S connectors, joints T-10 through T-12, are worked to 4 in. w.g. and the same notes say they are not recommended above that. The welded flange, joint T-21, is given no pressure ceiling at all; its note deals with flange width, edge welding and corner reinforcement. The companion angle, joint T-22, is plainly meant to go higher, because its bolt schedule sets 5/16 in. minimum bolts at 6 in. maximum spacing on the 4 in. w.g. class or lower and 4 in. maximum spacing on higher pressures, and a joint would not need a rule for higher pressures if it were never used there. Read all of this as selection limits from a construction standard rather than as Canadian law: SMACNA governs where a project specification or a provincial code adopts it.

Q25. Answer: A

Distortion in light gauge is shrinkage: the weld metal and the heat-affected zone contract as they cool and pull the panel out of plane. Two levers control it, put less heat in and take heat out faster. Short stitch welds distributed along the seam let each deposit cool before the next one goes down beside it, so the shrinkage forces are scattered instead of accumulating along one line, and a chill bar or chill plate clamped behind the joint draws heat out of the sheet. A strongback clamped across the joint holds it straight while this happens. The three wrong answers all add heat or add filler. Slowing travel speed raises the heat put into every millimetre of seam. Opening the root gap means more filler metal in the joint and therefore more metal shrinking as it freezes. Preheating a thin panel puts heat into the very part that is already distorting.

Q26. Answer: D

The Pittsburgh lock (also called Pittsburgh seam) is the standard longitudinal closing seam for rectangular ducts. One edge is formed into a pocket (the "Pittsburgh pocket"), and the mating flat edge is inserted and then clinched (hammered) closed. It provides a strong, airtight closure along the full length of the duct.

Q27. Answer: B

S1.21 gives the strap its own sentence: tie straps, used on positive pressure only, shall be 1 in. by 1/8 in. (25 by 3.2 mm) minimum galvanized steel, and the smallest edge shall face the airflow. Both halves matter. A strap works only in tension, and positive pressure is the mode that pulls the two reinforcements toward each other across the duct; under negative pressure the same tie goes into compression, where a 3.2 mm strap would simply buckle, which is why S1.19.5 offers rods, tubing, pipe or angles for negative pressure service and no strap at all. The orientation rule is about the airstream - turned edge-on, the strap presents 3.2 mm to the air instead of 25 mm. Uncoated steel is out. The same clause requires tie rods to be galvanized steel and all internal ties, whether rod, tube, pipe or angle, to be of material having the same nature and corrosion resistance as the duct wall, with concealed components neither subject to nor causing galvanic corrosion. The contrast is worth holding: S1.12 does allow reinforcement to be uncoated or galvanized steel, but that is the external member, not a tie living inside the airstream. Nor does a tie excuse the attachment of the reinforcement to the duct wall. S1.23 says the use of ties does not void the need to attach reinforcements to the duct wall; the only relief given is that where ties occur outside the duct, on two-sided or four-sided reinforcement at 4 in. w.g. positive and over, the attachment within two inches of the corner is not required. Ties are also spaced at even intervals of duct width not exceeding 60 in. (1524 mm) up to 120 in. width, and any hole made for tie rod passage is kept to minimum size and sealed. SMACNA is a construction standard, not Canadian law, and applies where a specification or an adopting code calls it up.

Q28. Answer: A

Sentence 6.3.2.10.(7) gives three requirements for an exhaust duct connected to laundry-drying equipment: it shall be independent of other exhaust ducts, accessible for inspection and cleaning, and constructed of a smooth corrosion-resistant material. Sentence (9) adds that such ducts or vents shall discharge directly to the outdoors. Every one of those requirements is about lint. Lint settles wherever the air slows or the wall is rough, so a ribbed or flexible wall packs with it, a duct shared with another system spreads it, a duct nobody can open is never cleaned, and lint at a dryer's operating temperature is the fuel for a duct fire. Tying the dryer into the washroom exhaust is the near miss because that arrangement is genuinely permitted elsewhere in the same Article: Sentence (12) lets exhaust systems serving rooms with water closets, urinals, basins, showers or slop sinks, and those serving rooms with residential cooking equipment, be interconnected at the inlet of an exhaust fan when suitable back pressure devices stop odours crossing over. Laundry-drying equipment is not in that permission. Discharging into an attic or roof space is not a discharge to the outdoors at all and dumps the dryer's moisture into the roof assembly. Where several dryers are vented collectively, Sentence (8) sets out the only route: one common exhaust duct vented by one central exhaust fan, an interlock that starts that fan when the laundry-drying equipment is in use, and makeup air provided to the space.

Q29. Answer: D

The seam line is where the two edges of the flat pattern come together when the blank is rolled or folded up into the fitting, and where the lock, groove or weld is then made. It is normally laid out on the shortest element of the pattern, the throat of an elbow for instance, so that the seam is as short as it can be and the least material and labour go into closing it. The other lines on the same pattern each do something different: the outline of the blank is the cut line, the stretchout or base line is the reference the element heights are measured from, and the bend lines mark where the metal folds. None of those three is the seam.

Q30. Answer: A

Plenum spaces contain energised conduit, communication cables, and the ceiling grid — contact with live wiring or stepping through the grid are key hazards requiring careful footing and awareness.

Q31. Answer: C

A notcher (corner notcher) cuts square or mitre notches at the corners of flat blanks. Without notching, the corner material would overlap or interfere when the box sides are bent up.

Q32. Answer: A

A cone is rolled by giving the sheet more curvature at one edge than at the other, and on a three-roll slip roll that is done at the rear roller: one end of it is raised (or lowered) so the roller sits at an angle to the front pair. The edge of the sheet running through the closer end takes the tighter radius and becomes the small end of the cone, and the edge at the open end takes the larger radius. The two front rolls are the pinch pair. They grip and drive the sheet and their gap is set to the material thickness, so setting one of their ends tighter would only pinch and mark the sheet at that end rather than changing the curvature, which is set behind them. Feeding the sheet at an angle changes where the metal enters, not how hard it is curved, and it makes the cone spiral away from the operator instead of tapering. There is no tapered die: a slip roll forms by roller position alone, which is why the same machine rolls any diameter within its range without tooling changes.

Q33. Answer: D

Compound fittings use combined methods: parallel line for developable flat/cylindrical faces and triangulation for non-developable transition zones.

Q34. Answer: C

SMACNA S1.15 requires duct sides 483 mm (19 in.) and over, of 20 gage (1.00 mm) or lighter, with more than 0.93 sq m (10 sq ft) of unbraced panel area, to be crossbroken or beaded, unless the duct is lined or externally insulated. Here the 600 mm sides are 23.6 in. wide and each unbraced panel measures 0.6 m by 2.4 m, or 1.44 sq m, so they qualify. The 400 mm sides are only 15.7 in. wide and are exempt on width alone, whatever their panel area. Crossbreaking and beading deal with out-of-flatness, sag and sheet flutter when the pressure is not enough to pull the sheet taut, and the standard says plainly that neither is a substitute for reinforcement - the reinforcement here is the rated transverse joint at 2.4 m spacing.

Q35. Answer: B

An eccentric reducer is an oblique cone. Its element lines still run to a single apex, because every cone has one, but the apex sits off to one side, so the elements are of unequal length. Each one must therefore be found in true length, by revolution or from a true-length diagram, before the pattern is struck. A right cone is the special case in which all the elements are equal, and that is what lets its pattern be swung with a single radius from one centre. Parallel line development suits shapes of constant cross-section, such as straight round pipe, not a taper. A concentric pattern cannot be rescued by trimming the small end either, because the offset changes the length of every element around the fitting.

Q36. Answer: C

Two subsections apply and both must be satisfied. BC OHS Regulation 12.116(1) states that a container which may have held a combustible substance must be thoroughly cleaned before any welding or burning operation is carried out on it. Subsection (2) then states that burning, welding or other hot work must not be done on any vessel, tank, pipe or structure, or in any place where the presence of a flammable or explosive substance is likely, until tests have been made by a qualified person to ensure the work may be safely performed and suitable safe work procedures have been adopted, including additional tests at intervals that keep the workers safe. Cleaning alone is therefore not enough; the atmosphere inside has to be proven, and proven again as the work goes on, because a warming drum keeps driving vapour out of residue that cleaning missed. Airing the drum open for a day is the near miss, and it is the practice that kills people. Solvent soaks into seam laps and sits under the rolled chime where nothing rinses it, so a drum can read clean, smell clean and still hold a vapour concentration in the flammable range; time passing proves nothing, and only a test by a qualified person does. Purging with oxygen is the dangerous inversion of a real technique. The recognised purges are an inert gas or filling the drum with water to drive the vapour space out; oxygen does the opposite, enriching the atmosphere so that anything ignitable burns far more violently. Keeping an extinguisher within reach is required at any welding location by section 12.126, but an extinguisher is a response after ignition, and a drum that lets go is an explosion rather than a fire to be put out.

Q37. Answer: D

SMACNA's notes for Figure 1-5, Type L-2 button punch snaplock, state that the seam may be used for 4 in. w.g. (1000 Pa) static or less, and that screws must be added at the ends on all duct of the 4 in. w.g. class and at the ends of 3 in. w.g. class duct over 48 in. wide. The same notes recommend a 5/8 in. pocket depth for 20, 22 and 24 gauge and warn against the seam on aluminum and other soft metals. Note what the standard does not do: it does not assign one longitudinal seam to each pressure class. Pittsburgh, grooved and standing seams are listed alongside the snap lock, and the seam is selected for the material, the pressure classification and the rest of the construction detail. SMACNA is a construction standard rather than Canadian law; it governs where a specification or a provincial code adopts it, and the Red Seal Occupational Standard for the trade names it in sub-task B-7.02.

Q38. Answer: D

SMACNA's text on sealing says it directly: contractors using slip and drive connection systems must control connector length and notch depth on rectangular duct ends, and failure to do so will compromise seal effectiveness. The reason is the corners. A drive cut long, or an end notched deeper than it needs to be, leaves an opening at the corner where the S-cleat, the drive and the two duct ends all meet, and that opening is the hardest place on the joint for a sealant to bridge. Liquid sealants are the type used on slip joints because they fill the small space between overlapping metal, but they are 30 to 60 percent volatile solvent and shrink considerably as they dry, so where the metal clearance exceeds 1/16 in (1.6 mm) several applications may be needed just to fill the void left behind. A corner gap opened by careless notching is far wider than that. Sealant brand and application temperature matter to the product's own performance but they are not what SMACNA names here, sheet gauge and pressure class govern the construction rather than the sealing of the corner, and rivet dimensions belong to a fastened lap rather than to a slip-and-drive joint, which is not riveted.

Q39. Answer: C

Spot weld nugget size is primarily controlled by current (amperage). Higher current → larger nugget. Weld time and electrode force also affect nugget; current is the dominant variable.

Q40. Answer: B

Decanting is what triggers the workplace label. CCOHS lists three situations that call for one: a hazardous product made at the workplace and used there, a supplier label that is lost or illegible, and a hazardous product decanted, that is transferred or poured, into another container. In general a workplace label carries the product name matching the safety data sheet, safe handling precautions, which may be given as pictograms or other information taken from the supplier label, and a reference to the safety data sheet. CCOHS writes that last element as a reference to the sheet if available, and for a supplier-labelled hazardous product like this sealant the sheet is available. Two situations let a shop off the full label: the product is poured out and used immediately, or it stays under the control of the person who decanted it, meaning that person is the only one who will use it and it is used during one shift. Even then the container must still be identified with the product identifier. Neither situation covers a can left on the bench all week for the whole crew, and CCOHS says plainly that a full workplace label is required once the product is not used right away or more than one person will be in control of it. The supplier address and telephone number and the signal word are supplier label elements, put on the pail by the Canadian manufacturer or importer; the employer's workplace label is not obliged to reproduce them, and none of them tells the next worker how to handle what is in the can. Copying the pictogram and signal word off the pail leaves the can with no product name at all, which is the one thing even an exempt container must show. Workplace label requirements are set by the provincial or territorial jurisdiction, or by the Canada Labour Code in a federally regulated workplace, so confirm the exact wording with your own jurisdiction.

Q41. Answer: C

The stretch-out, or girth, of a rectangular duct is the distance right around its cross-section: twice the width plus twice the depth, so $2 \times (400 + 250) = 1300$ mm. That figure covers the finished surface of the duct and nothing else. The Pittsburgh lock is made from metal that is not part of that finished perimeter. The pocket rolled on one edge and the plain folded edge that enters it are both extra metal, so their allowances are added to the girth rather than found inside it, and the same is true of any hem or flange added at the ends. Taking the lock allowance out of the girth runs the calculation backwards and leaves the section short by the depth of the lock, which cannot be made up once the sheet is cut. Halving the girth to 650 mm would be right only if the section were made from two L-shaped halves seamed at two corners, which is a different construction from the one the question describes. The Red Seal Occupational Standard puts the rule in the performance criteria for pattern development: the pattern points are connected, and allowances for seams and edges are added.

Q42. Answer: B

Two things have to be right, not one. The gap between the rest and the wheel must be small, within 3 mm (1/8 in.), and the rest must sit with its edge at or above the centreline of the wheel. A wide gap is what makes a grinder dangerous: the wheel turns down toward the rest, so a chisel or a small part can be dragged into the opening and wedged between the rest and the wheel, snatching it out of the hands and, at worst, breaking the wheel. Setting the rest low has the same effect for a different reason, since the work is then presented under the centre of the wheel where the rotation drives it downward into the gap instead of back against the rest. That is why the WorkSafeBC OHS Regulation asks for both conditions in one sentence and why CCOHS pairs the 3 mm rest setting with a tongue guard adjusted to leave about 6 mm (1/4 in.) of wheel exposed at the top. Swinging the rest out of the way to lay a long piece on removes the support the tool is meant to have and leaves the work held only in the hands. Note also that the rest is adjusted with the wheel stopped, never while it is turning, and that a straight wheel is used on its face and not on its side.

Q43. Answer: A

A mallet with a rawhide, hide, plastic or hardwood face delivers the blow without a hard, narrow steel face driving into the metal. On light gauge sheet a steel hammer face does two things at once: it thins and stretches the metal locally, so the panel is left dished and the fold line wanders, and it dents through the zinc, leaving a bright break in the coating that the sealant or paint then has to cover. A mallet spreads the blow, closes the fold and leaves the surface flat. That is why the Red Seal tool list for this trade carries mallets alongside raising, setting, ball peen and riveting hammers. The steel hammers have their own work - driving another tool such as a hand groover, setting rivets against a dolly, dressing a seam that is already formed - but they are not swung onto an open panel face. A mallet is generally lighter than a steel hammer of the same head size, not heavier, so fewer blows is not the reason. Sparking is not the issue either: duct sealants are applied cold and mallets are not chosen here as non-sparking tools. And no rule bars steel hammers from work on coated sheet; they are in constant use on it, driving groovers and setting rivets, just not swung onto an open panel face.

Q44. Answer: D

True length is the hypotenuse of a right triangle whose base is the plan length and whose height is the vertical rise: $TL = \sqrt{(400^2 + 300^2)} = \sqrt{(160000 + 90000)} = \sqrt{250000} = 500$ mm. This 3-4-5 relationship is the basis of the true-length diagram used in triangulation.

Q45. Answer: B

VAV boxes contain a motorised modulating damper controlled by the zone thermostat. They vary airflow from maximum to minimum (or zero) to maintain setpoint temperature.

Q46. Answer: C

A square elbow without turning vanes creates significant turbulence, high pressure drop, and noise as air separates from the outer wall. Turning vanes (sheet metal blades inside the elbow) guide airflow in a smooth arc, reducing separation, pressure drop by up to 75%, and noise. Required by many specifications for critical systems.

Q47. Answer: C

The grooved seam (or rolled groove seam) is used on round ductwork — particularly the spiral seam formed by a spiral pipe machine — and for connecting two curved edges. Both edges are formed into matching grooves and then rolled together to lock. It creates a smooth, mechanically tight seam without exposed sharp edges.

Q48. Answer: D

A box and pan brake, also called a finger brake, has a segmented upper beam. The fingers can be removed and rearranged so that the beam clears flanges already turned up on the workpiece, which is what lets the second, third and fourth sides of a box or pan be bent without fouling the sides already formed. Individual fingers are combined to make up whatever width the next bend needs. Rolling sheet into cylinders is the slip roll's job, cutting blanks to length is the squaring shear's, and a Pittsburgh pocket is rolled on a lock former.

Q49. Answer: C

Solid rivet installation: 1) Drill or punch the hole to a close fit — only a few thousandths of an inch over the rivet shank, so the shank fills the hole as it is driven and the joint carries its load in shear. A #30 drill (0.1285 in.) for a 1/8 in. rivet is about 0.0035 in. oversize. 2) Insert the rivet. 3) Support the factory head with a bucking bar (dolly) held firmly against it. 4) Drive the rivet shank with a pneumatic rivet gun or hammer to upset (flatten) the shank into a shop head. The resulting compression holds the sheets together. A loose, oversize hole lets the shank bend rather than fill, and the joint works loose.

Q50. Answer: D

When a bend is released, elastic stress partially straightens the metal. To achieve the required final angle, the sheet must be overbent slightly. The overbend angle depends on material and thickness.

Scored under 70%?

The real exam draws from a much wider pool. The **308A Sheet Metal Worker Complete Question Bank** gives you every question on our platform with full explanations and key-concept summaries — built to close exactly the gaps this mock exposed.

Get the 308A Sheet Metal Worker Complete Question Bank →

Or keep practising free online at redsealquiz.ca/308a.html

© 2026 · redsealquiz.ca/308a.html · For personal study use only. Independent, unofficial study resource — not affiliated with or endorsed by the CCDA, the Red Seal Program, or any apprenticeship authority. See "About This Material".